



MMS050 Finite Element Simulation In Design

Hand in Assignment 3

Ramkumar Huleppa Munavalli
Email : huleppa@student.chalmers.se

Chalmers University of Technology
MMS050 Finite Element Simulation In Design

December 18, 2025

Contents

1	Task 1: Buckling of the water tower	2
1.1	SubTask	2
1.2	SubTask	4
2	Task 2: Buckling of a stiffened panel	5
2.1	SubTask	5
2.2	SubTask	8
2.3	SubTask	10
3	Task 3: Geometric Nonlinearity	12
3.1	SubTask: Linear Analysis	12
3.2	SubTask: Nonlinear Analysis	13
4	Task4:Modal analysis	14
4.1	SubTask	14
4.2	SubTask	15
5	Task 6: Thermo-mechanical analysis of a concrete slab	17
5.1	SubTask	17
5.2	SubTask	19

1 Task 1: Buckling of the water tower

1.1 SubTask

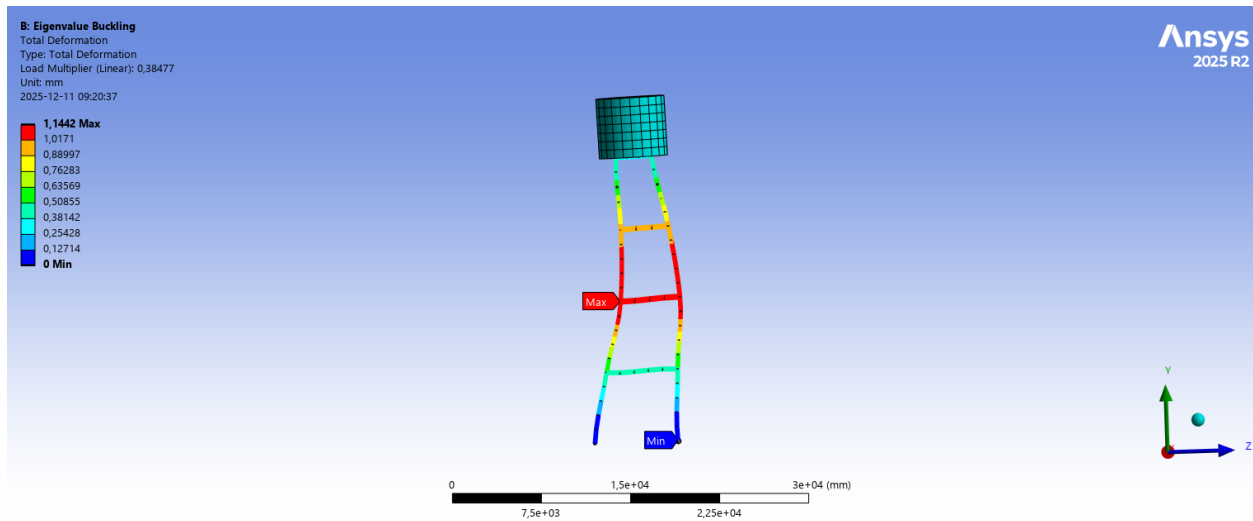


Figure 1: Mode 1 Total deformation

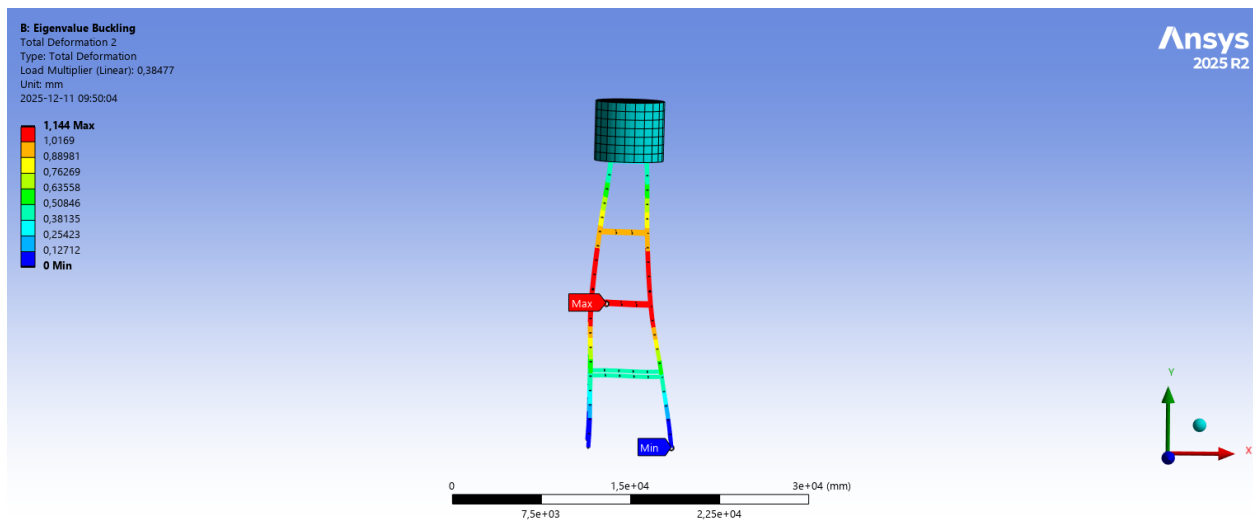


Figure 2: Mode 2 Total deformation

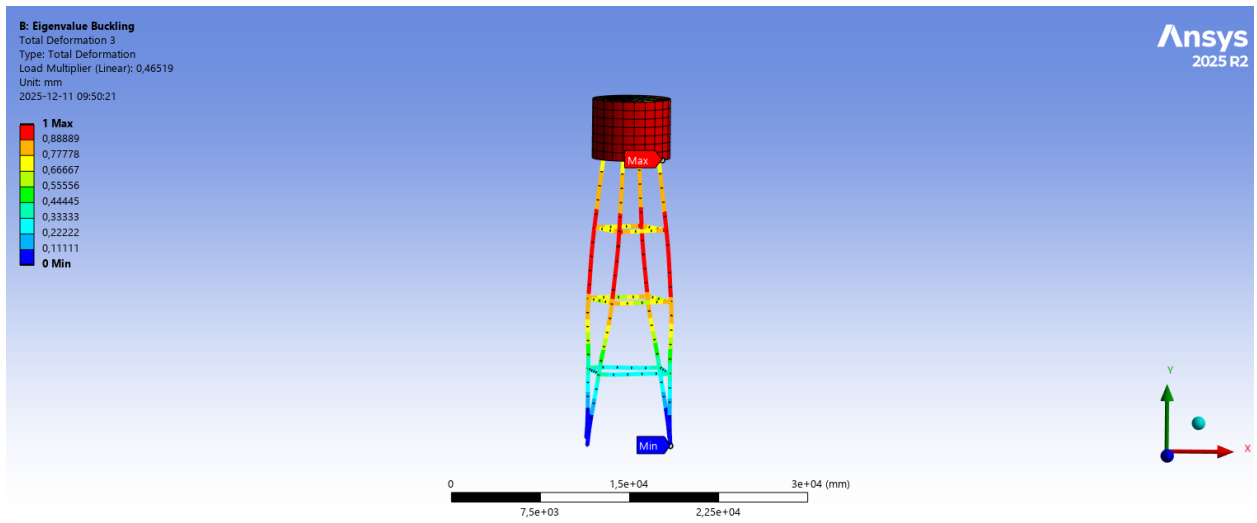


Figure 3: Mode 3 Total deformation

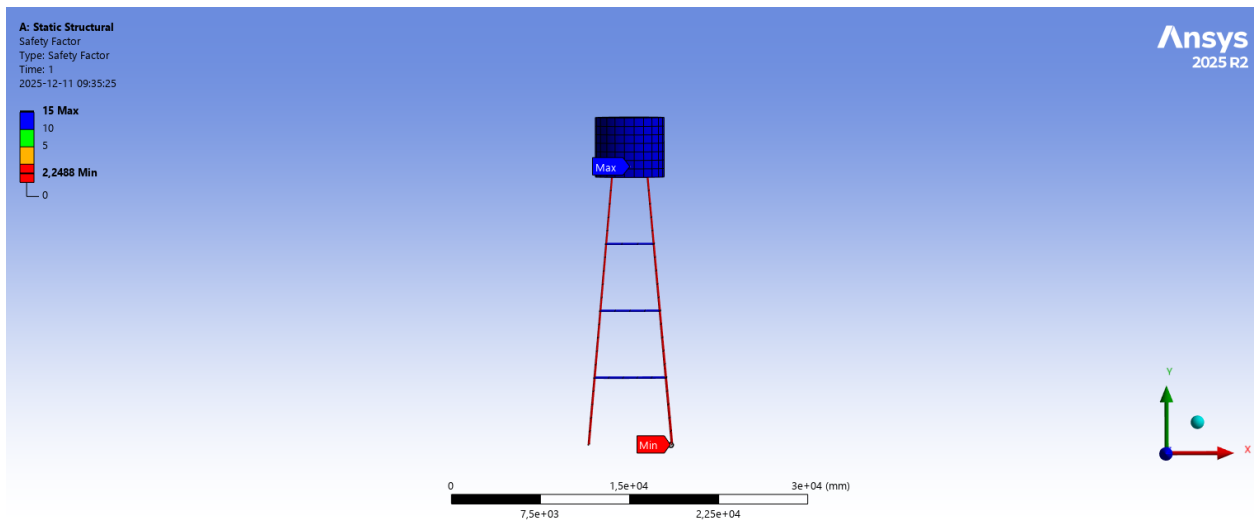


Figure 4: Factor of Safety

The cross section of the frame is $R_0 = 60\text{mm}$ and $R_i = 40\text{mm}$. The Lowest load factor is 0.38477 for the Water Tank. The factor of safety is around 2.2488.

1.2 SubTask

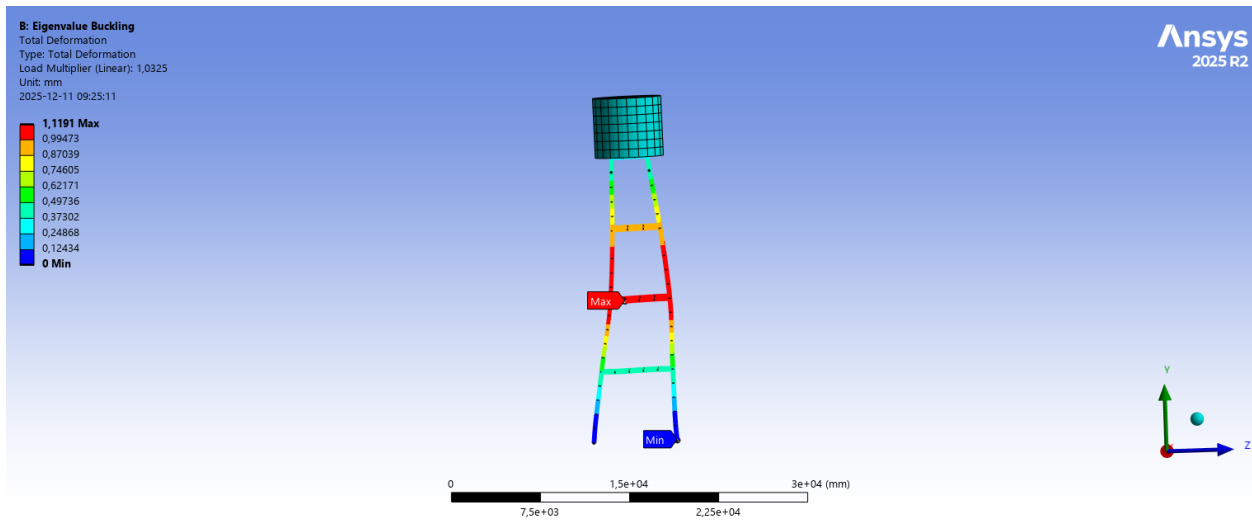


Figure 5: Mode 1 Total deformation

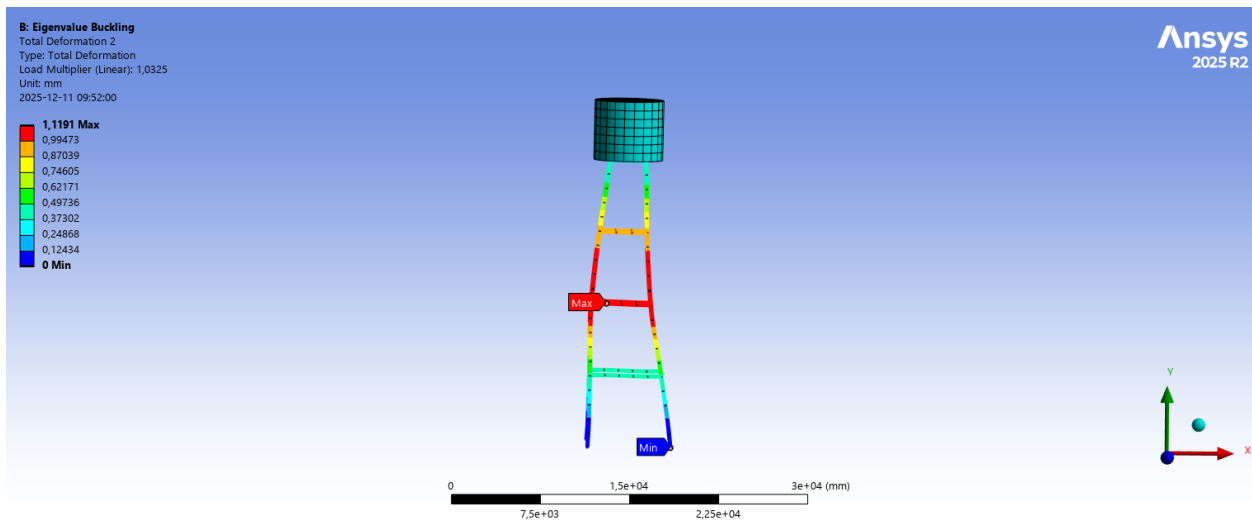


Figure 6: Mode 2 Total deformation

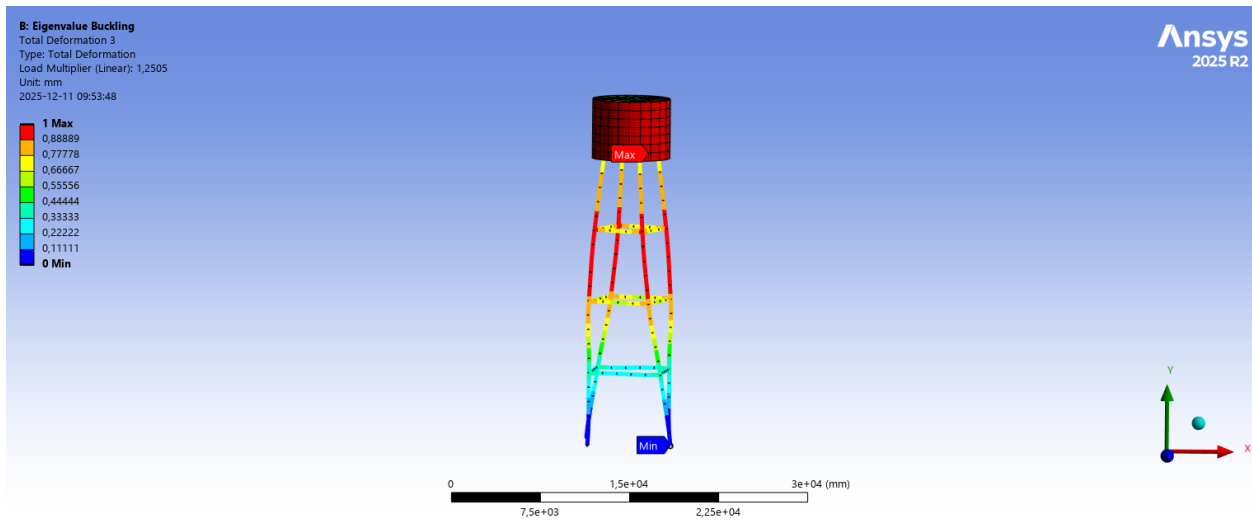


Figure 7: Mode 3 Total deformation

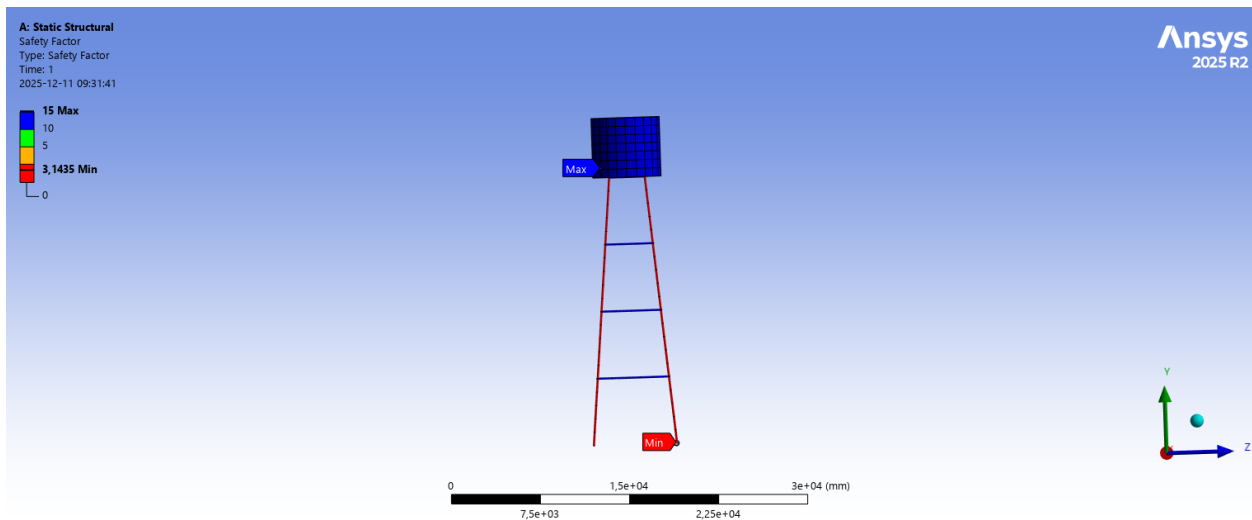


Figure 8: Factor of Safety

The cross section of the frame is $R_0 = 80\text{mm}$ and $R_i = 60\text{mm}$. The Lowest load factor is 1.0325 for the Water Tank. The factor of safety is around 3.1435..

2 Task 2: Buckling of a stiffened panel

2.1 SubTask

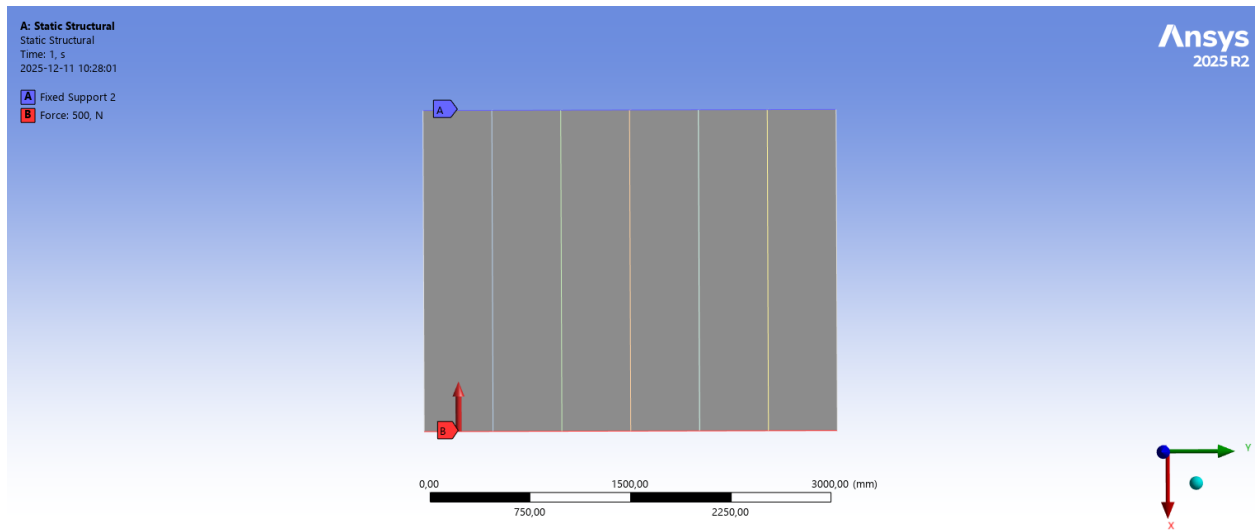


Figure 9: Boundary condition

The model is a simply supported, while a compressive force of 500N is applied. The opposite end of the model is fixed.

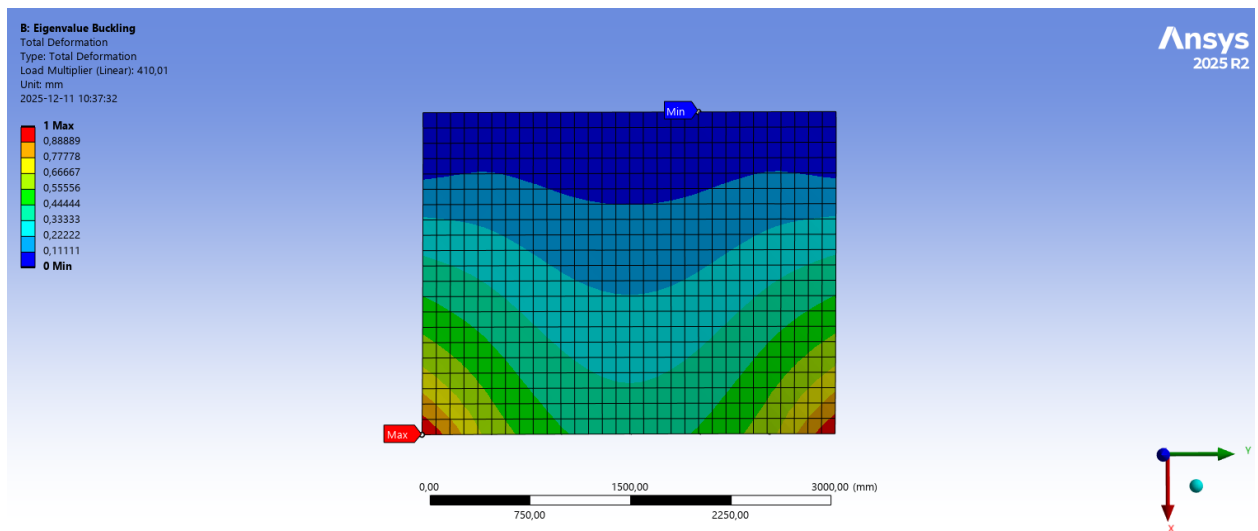


Figure 10: Mode 1 Total deformation

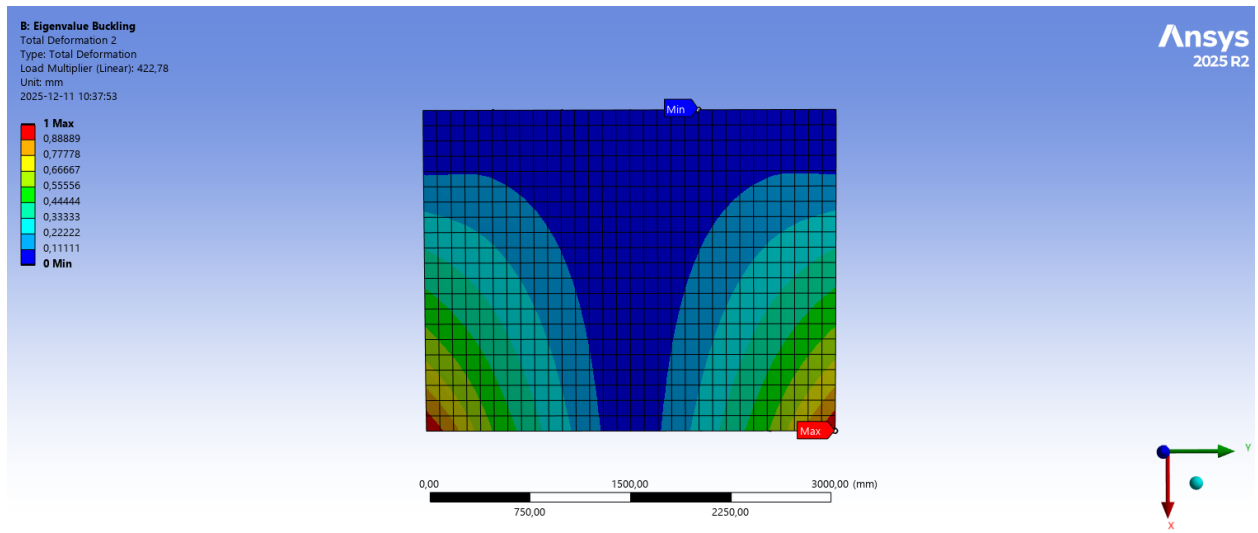


Figure 11: Mode 2 Total deformation

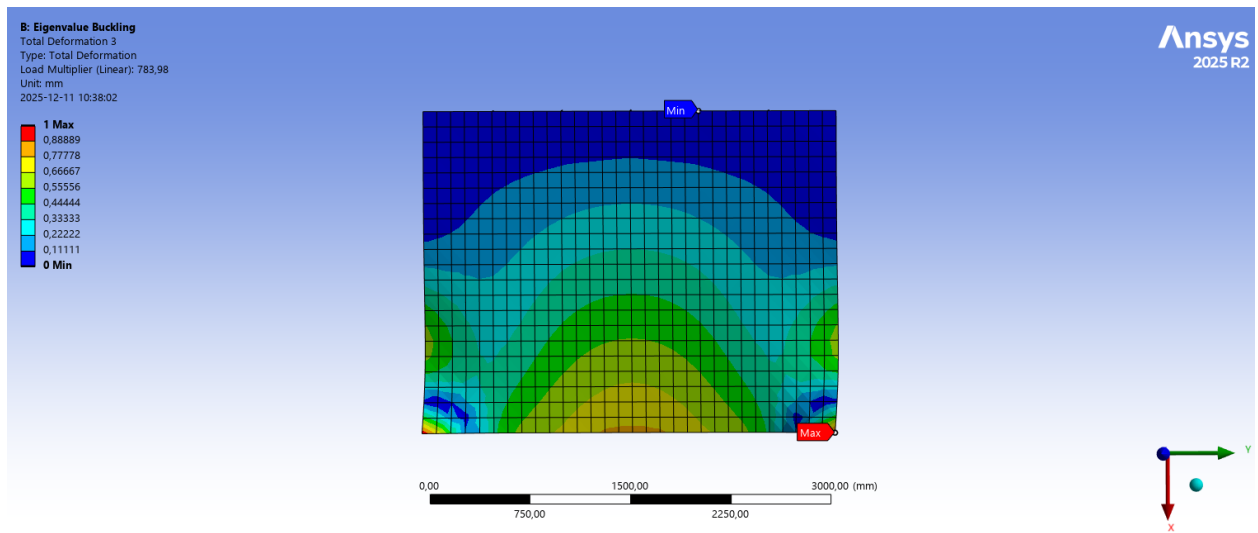


Figure 12: Mode 3 Total deformation

Mode	Load Multiplier	Critical Load(N)
1	410.01	205005
2	422.78	211390
3	783.98	391990

Table 1: Critical load for 3 lowest buckling modes

2.2 SubTask

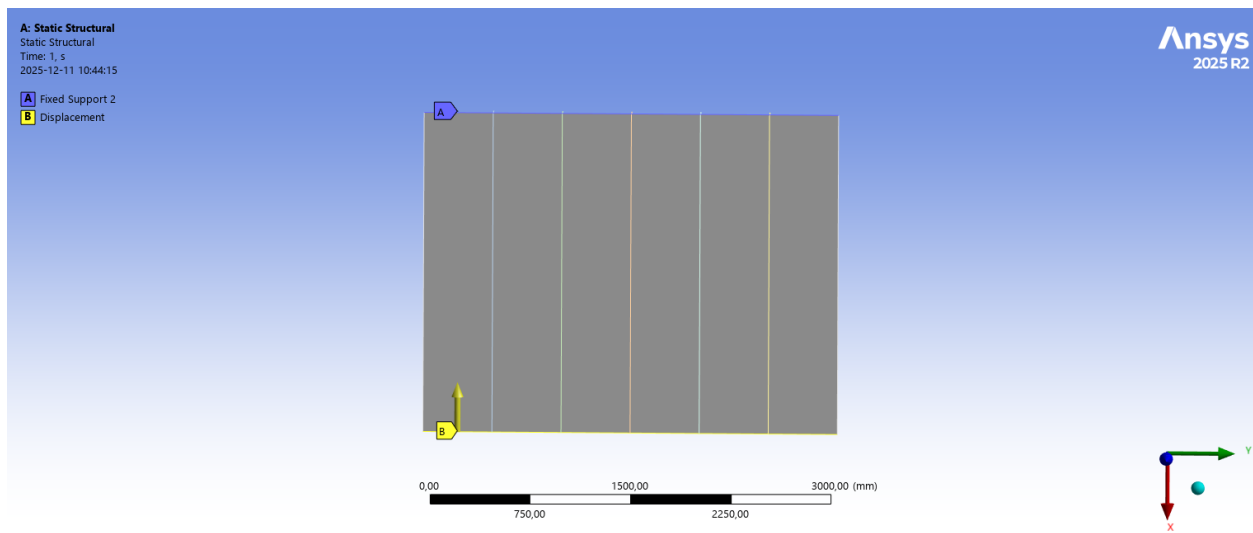


Figure 13: Boundary condition

The buckling load is replaced by nodal displacement of 1mm. The opposite end of the model is fixed.

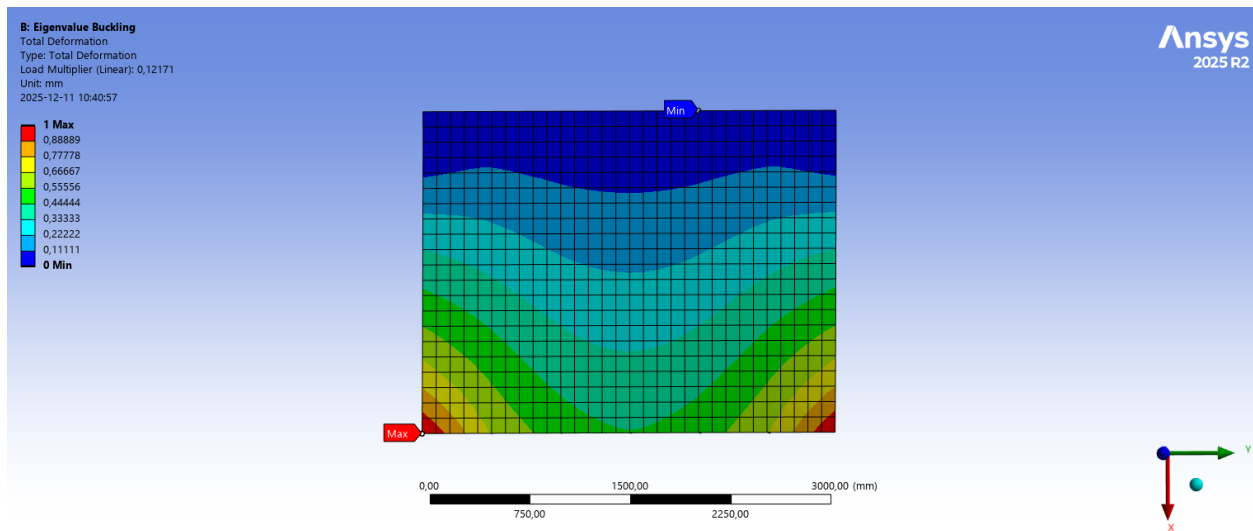


Figure 14: Mode 1 Total deformation

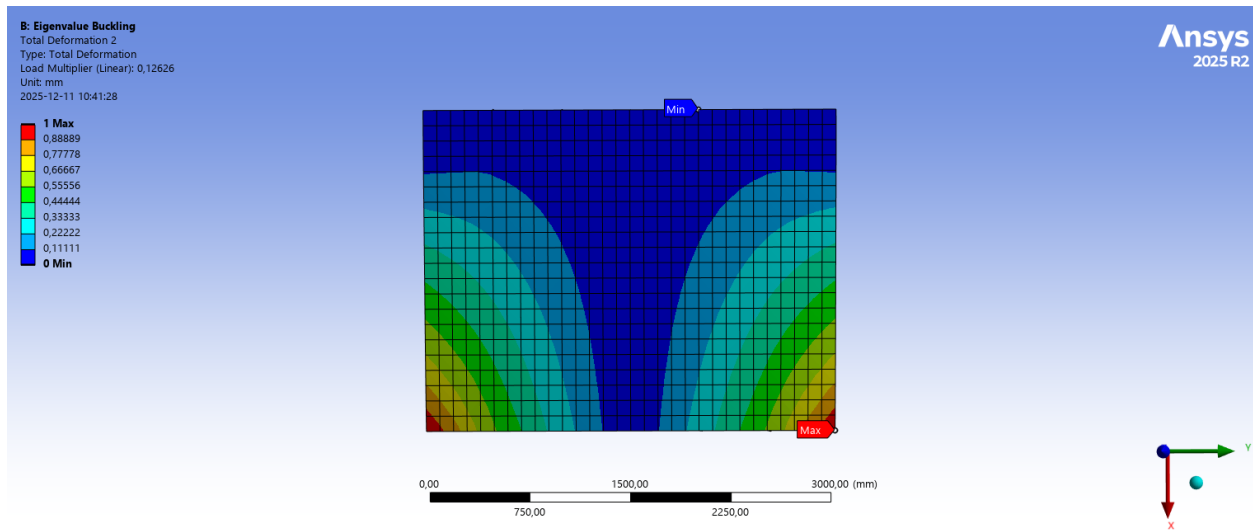


Figure 15: Mode 2 Total deformation

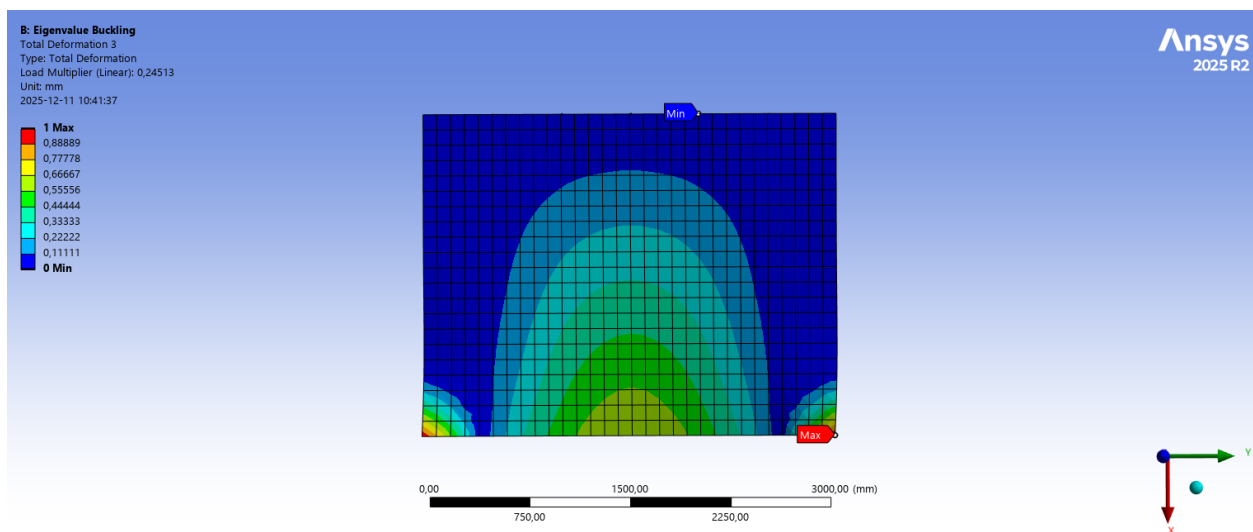


Figure 16: Mode 3 Total deformation

The buckling modes exhibit significantly lower load limits, as it is already close to its critical buckling condition. Even a small additional increase in the applied compressive load may trigger instability and lead to buckling.

2.3 SubTask

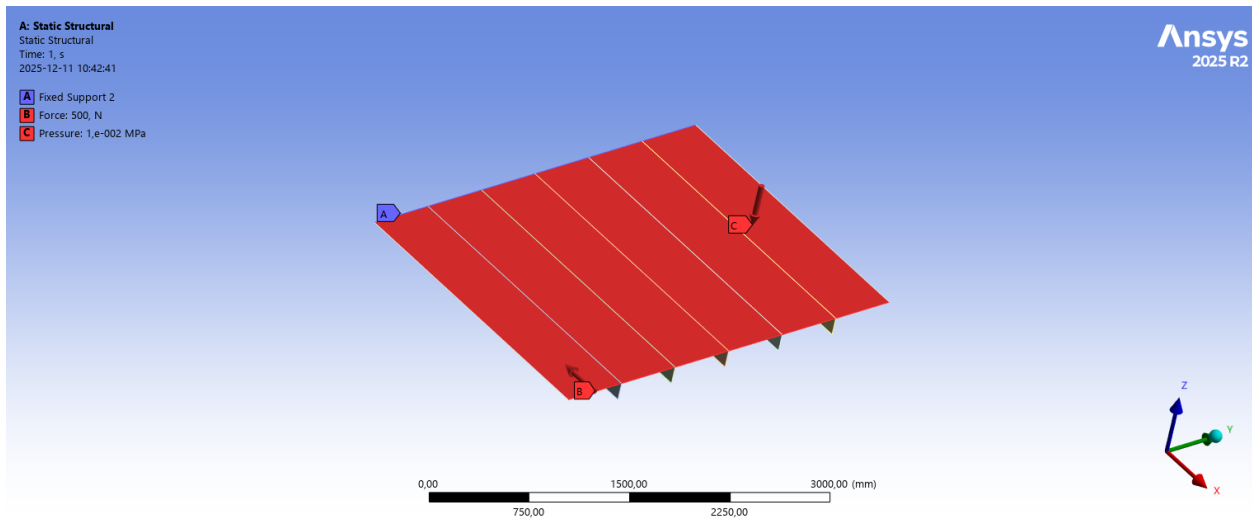


Figure 17: Boundary condition

The compressive force of 500N is applied. The opposite end of the model is fixed and the panel is subjected to lateral pressure in -z direction of 10Kpa.

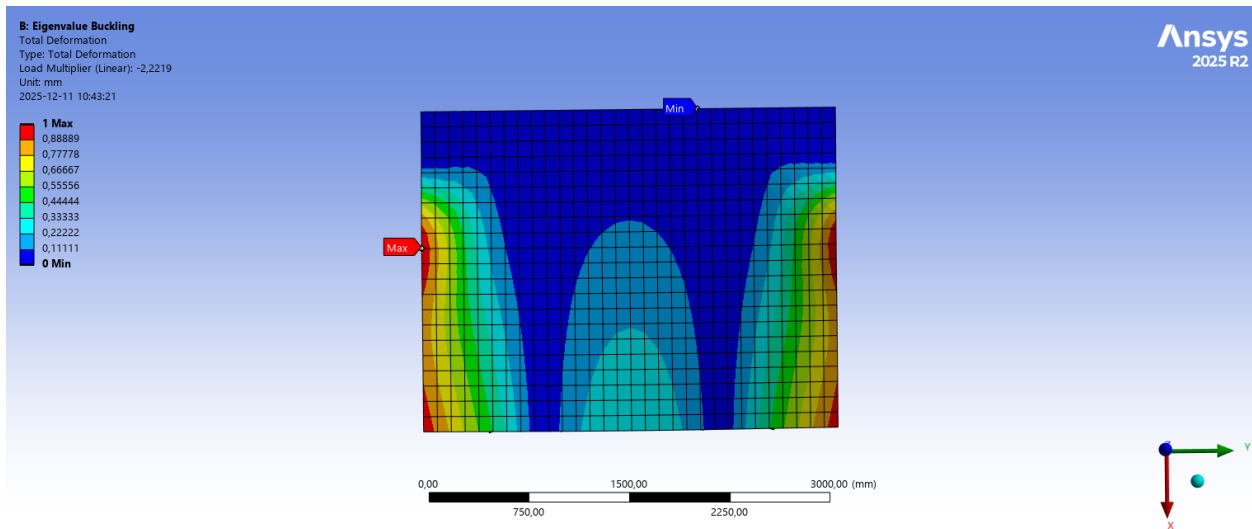


Figure 18: Mode 1 Total deformation

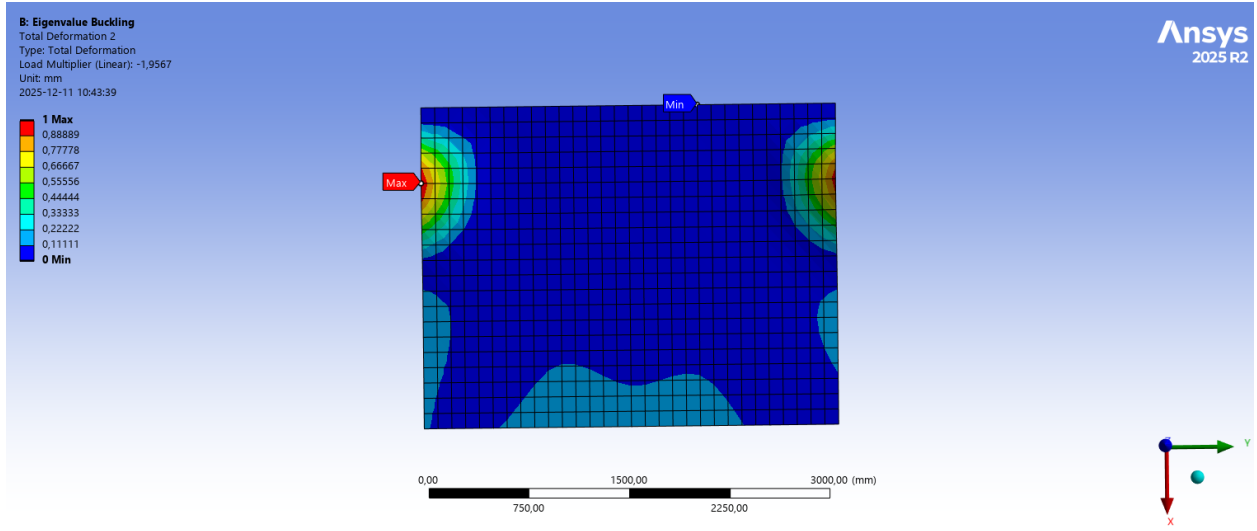


Figure 19: Mode 2 Total deformation

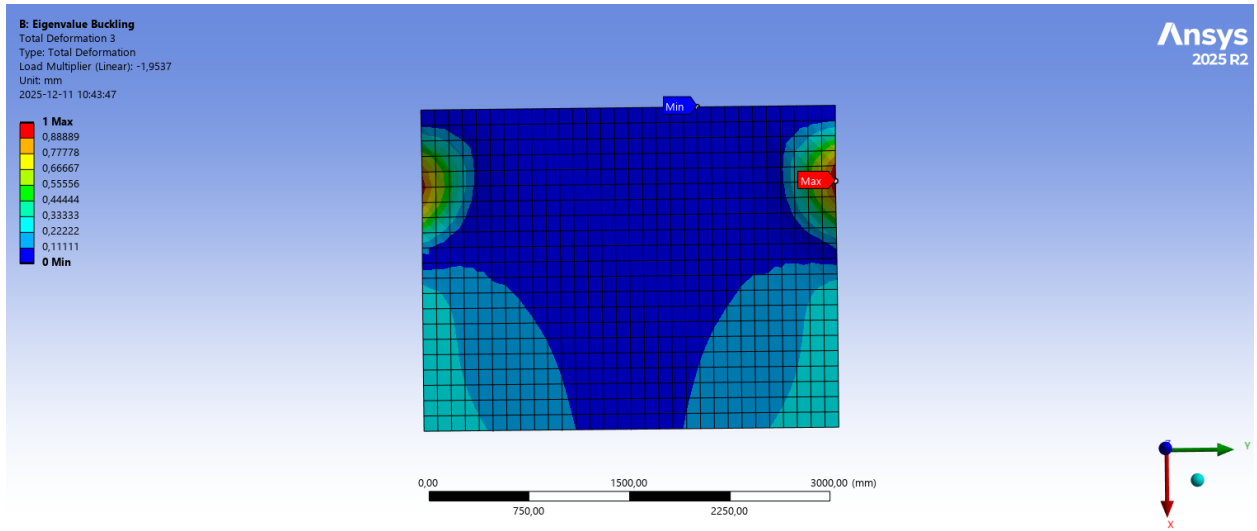


Figure 20: Mode 3 Total deformation

Mode	Load Multiplier	Critical Load(N)
1	-2.2219	-22219
2	-1.9567	-19567
3	-1.9537	-19537

Table 2: Critical loads and load multiplier for lateral pressure

In this case, the results show negative values, which means the buckling modes would occur under a tensile load rather than a compressive one. Compared to the previous case, which had almost no applied pressure, this setup is less likely to buckle under compression and more prone to buckling when subjected to tension.

3 Task 3: Geometric Nonlinearity

3.1 SubTask: Linear Analysis

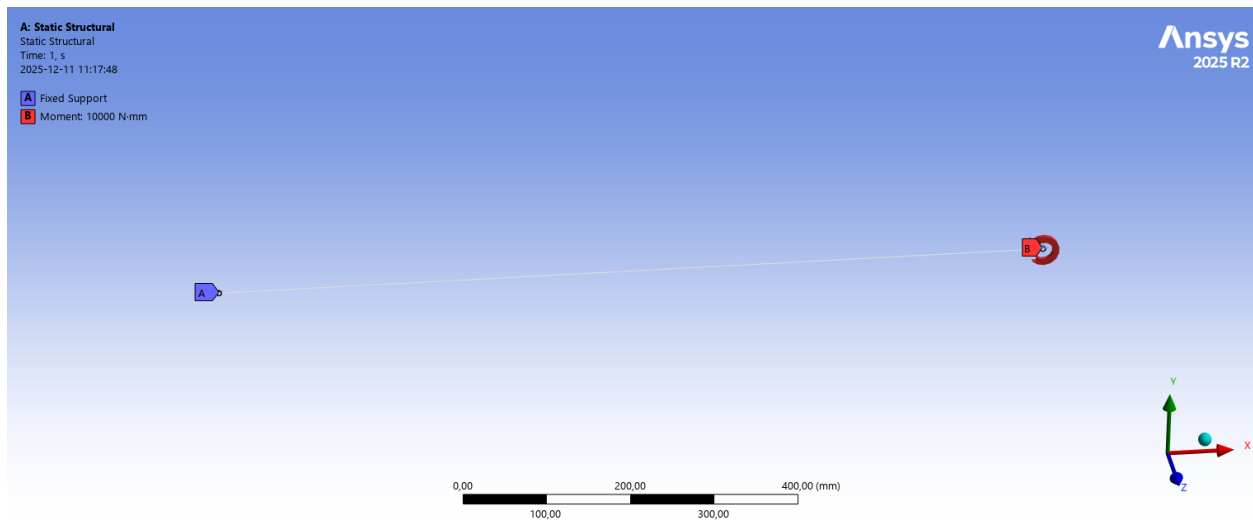


Figure 21: Boundary Condition

The fixed support is applied to one end of the beam and the moment of 10000Nm around z axis at the other end of the beam.

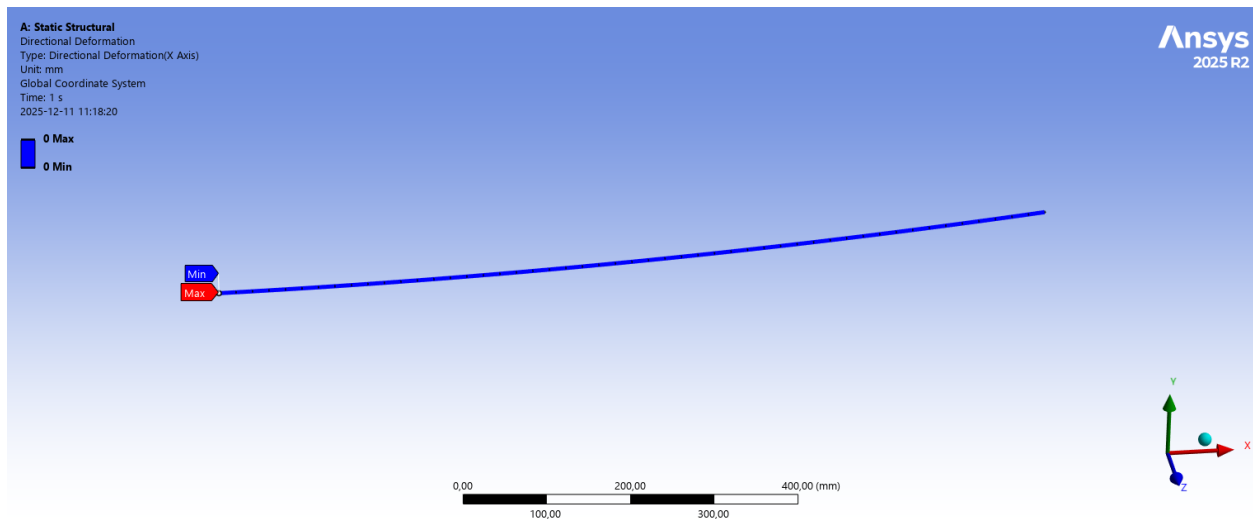


Figure 22: Direction deformation in X-axis

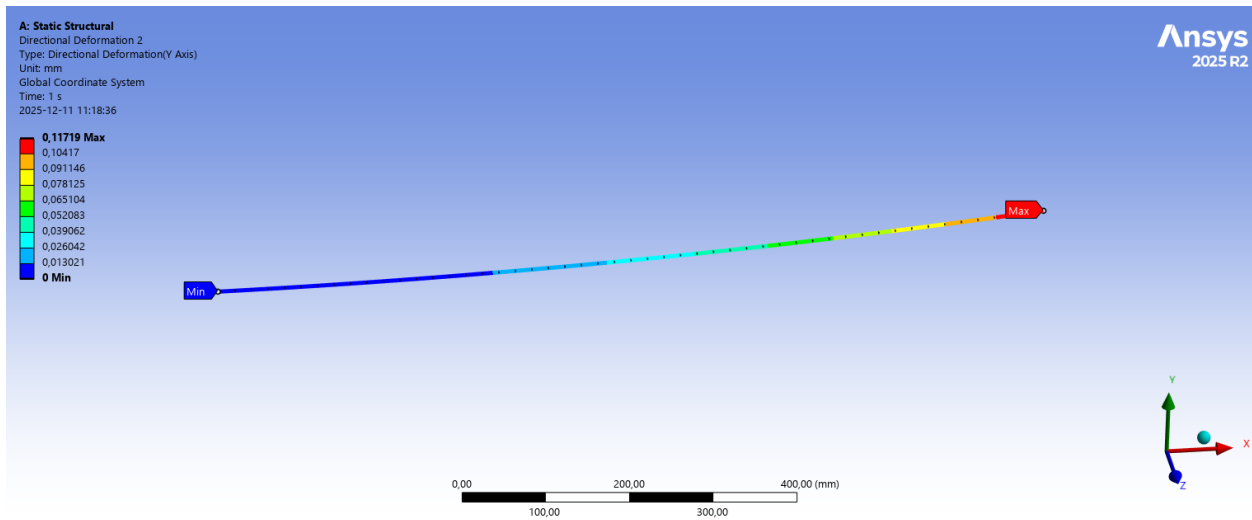


Figure 23: Direction deformation in Y-axis

Linear FE predicts zero axial X deformation for pure end moment, as bending doesn't stretch the centroidal axis. Under a pure end moment about the z-axis, the beam is expected to bend in a constant-curvature shape, with the free end rotating and deflecting in the transverse direction. Core linear FE assumption: small displacements/rotations vs. beam dimensions, with constant undeformed stiffness matrix ignoring geometric changes.

3.2 SubTask: Nonlinear Analysis

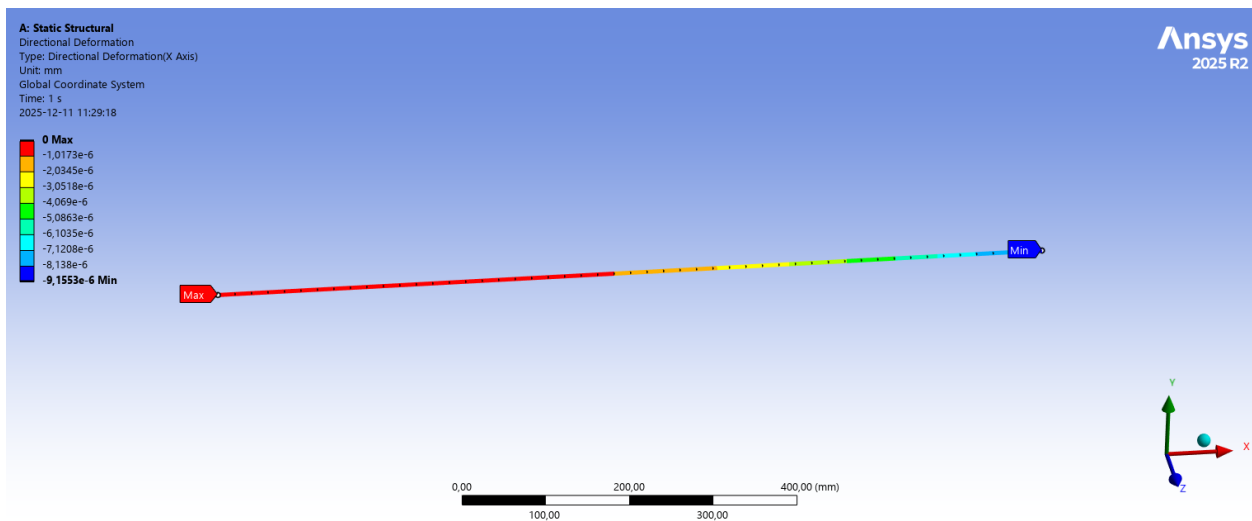


Figure 24: Nonlinear analysis direction deformation in X-axis

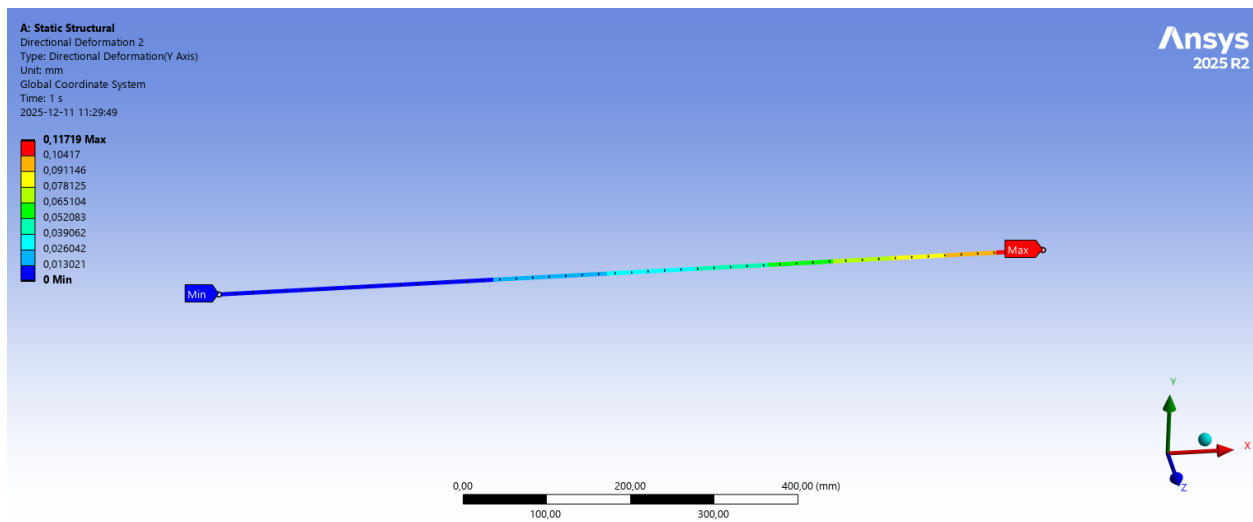


Figure 25: Nonlinear analysis direction deformation in Y-axis

Nonlinear analysis differs by updating the stiffness matrix iteratively for geometric nonlinearity, yielding smaller Y deformation, possible minor X deformation.

4 Task4:Modal analysis

4.1 SubTask

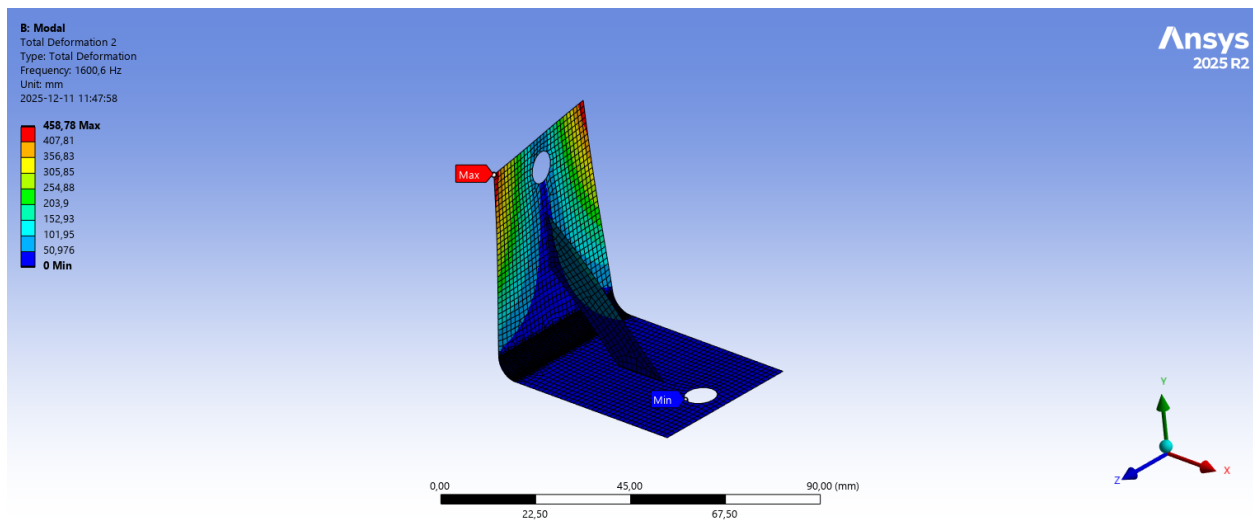


Figure 26: Mode 1 total deformation for load case

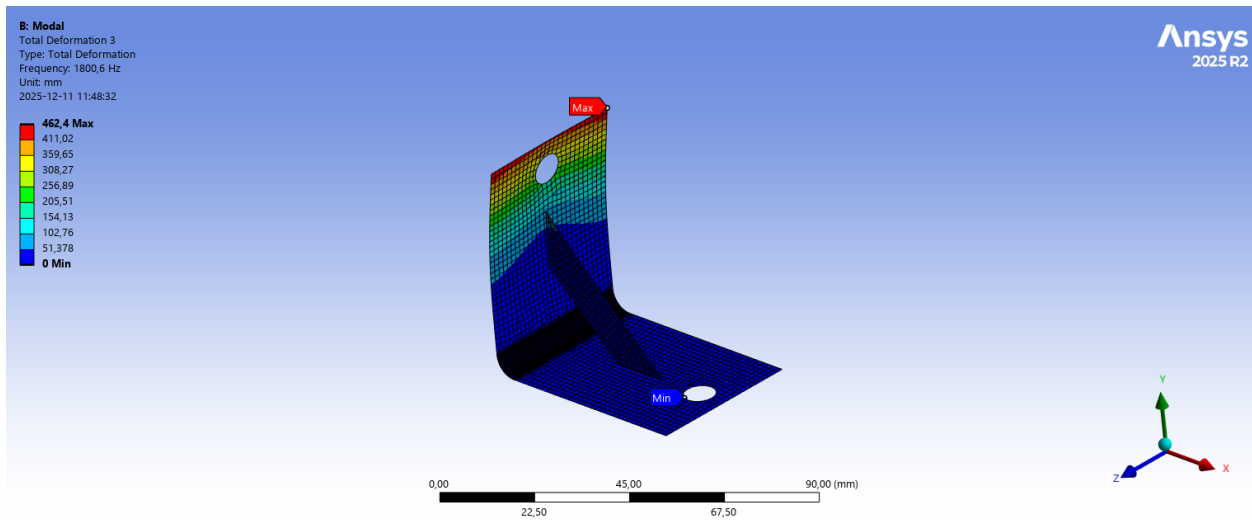


Figure 27: Mode 2 total deformation for load case

Fig 26 and Fig 27 are mode 1 and 2 total deformation. The Mode shapes are at 1960.7Hz and 1966.3 Hz.

4.2 SubTask

Outline of All Parameters			
	A	B	C
1	ID	Parameter Name	Value
2	Input Parameters		Unit
3	Static Structural (A1)		
4	P1	SYS/Gusset Thickness	1,5
5	P2	SYS/Bracket Thickness	3
*	New input parameter	New name	New expression
7	Output Parameters		
8	Static Structural (A1)		
9	P3	Geometry Mass	0,10409
10	P4	Equivalent Stress Maximum	1,1085
11	Modal (B1)		
12	P5	Total Deformation 2 Reported Frequency	2232,2
*	New output parameter		New expression
14	Charts		
15	Parameter Chart 0		
16	Parameter Chart 1		

Table of Design Points					
	A	B	C	D	E
1	Name	P1 - SYS/Gusset Thickness	P2 - SYS/Bracket Thickness	P3 - Geometry Mass	P4 - Equivalent Stress Maximum
2	Units	mm	mm	kg	MPa
3	DP 0 (Current)	1,5	3	0,10409	1,1085
4	DP 1	1,2	1,7	0,060117	3,3725
5	DP 2	1	2	0,069392	2,4957
6	DP 3	0,8	2,2	0,07536	2,1053
7	DP 4	0,8	1,7	0,058821	3,4622
*					

Figure 28: Table of design points

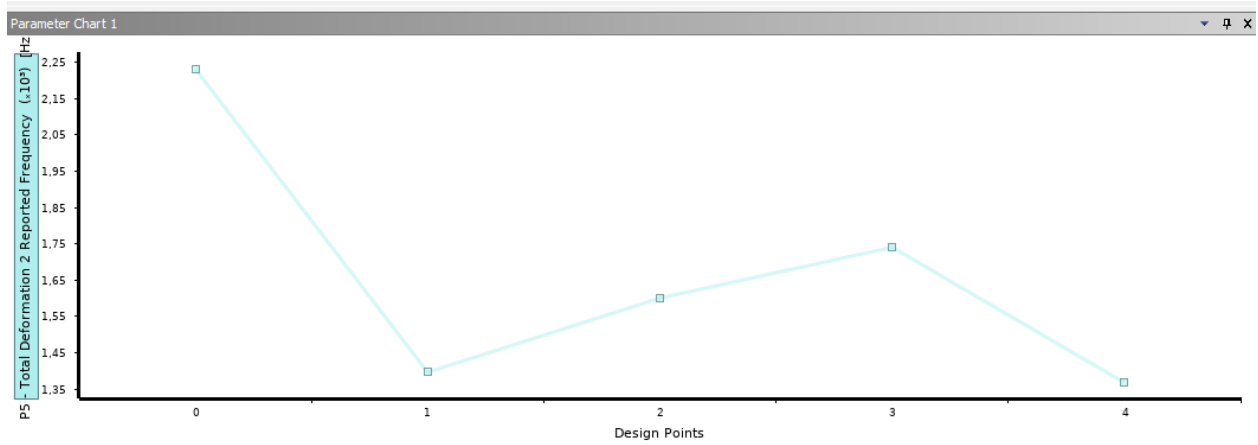


Figure 29: The frequency of the lowest eigenmode

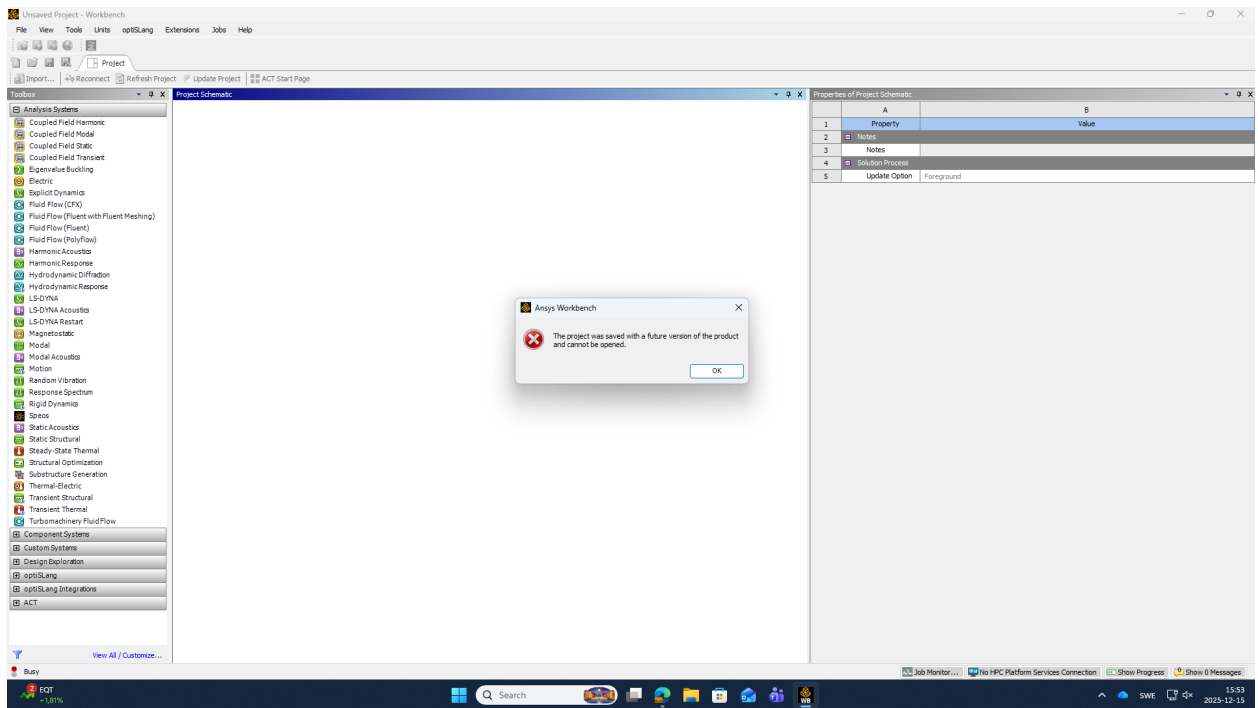


Figure 30:

I accidentally saved the model in ansys 2025 R2 version, for to add figures, i cannot open the model in ansys 2024 R2 version.

5 Task 6: Thermo-mechanical analysis of a concrete slab

5.1 SubTask

Outline of All Parameters

	A	B	C	D
1	ID	Parameter Name	Value	Unit
2	Input Parameters			
3	Steady-State Thermal (A1)			
4	P2	Convection Ambient Temperature	20	C
*	New input parameter	New name	New expression	
6	Output Parameters			
7	Steady-State Thermal (A1)			
8	P1	Temperature 2 Average	18,387	C
*	New output parameter		New expression	
10	Charts			

Properties: No data

	A	B
1	Property	Value

Table of Design Points

	A	B	C	D	E
1	Name	P2 - Convection Ambient Temperature	P1 - Temperature 2 Average	☐ Retain	Retain
2	Units	C	C		
3	DP 0 (Current)	20	18,387	☒	☒
4	DP 1	25	21,065	☐	
5	DP 2	30	23,743	☐	
6	DP 3	35	26,421	☐	
7	DP 4	40	29,099	☐	
8	DP 5	41,8	30,064	☐	
*				☐	

Chart: No data

Job Monitor...

No HPC Platform Services Connection

Show Progress

Show 1 Messages

Figure 31: Table of temperature

For the different water temperature, the surface temperature is been recorded.

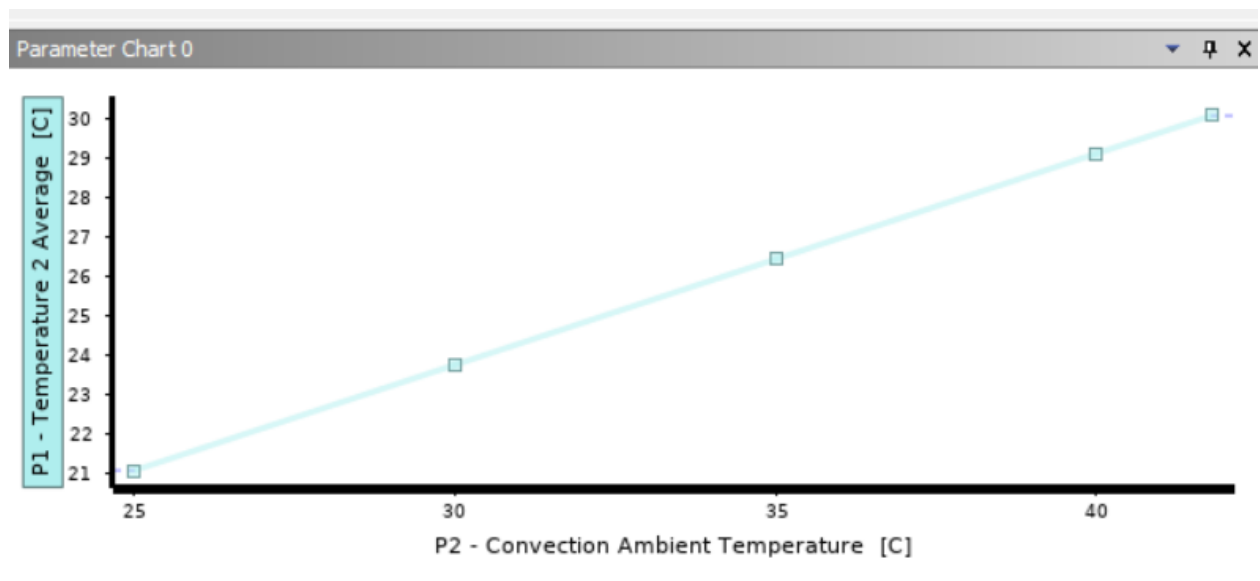


Figure 32: Convection ambient temp(C) vs Temperature(C)

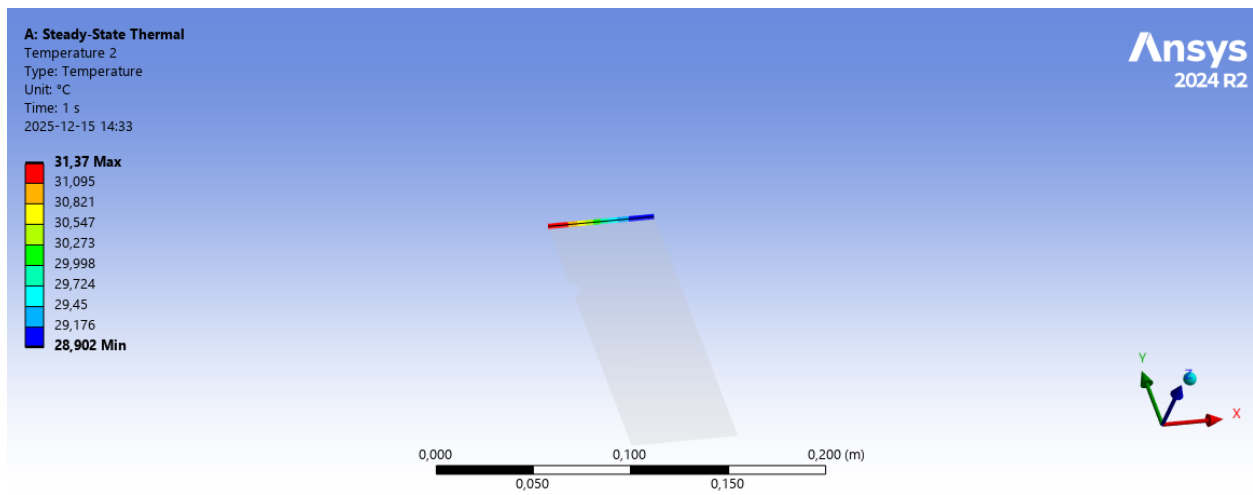


Figure 33: Temperature of top surface

For input water temperature of 41,8 C, the output surface temperature is 31,37 C.

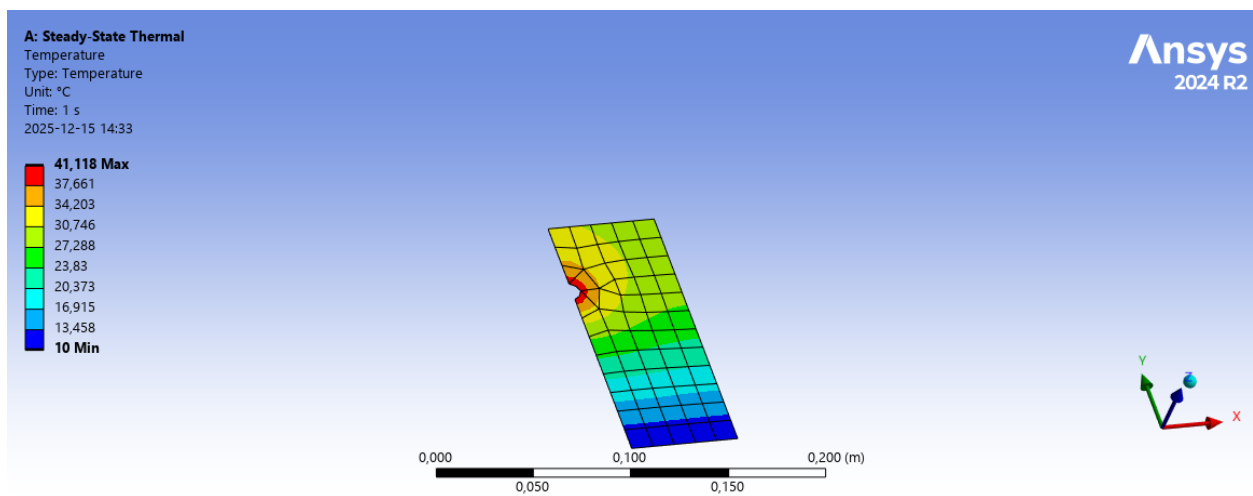


Figure 34: Temperature at the hole

The maximum temperature at the hole is 41,118 C

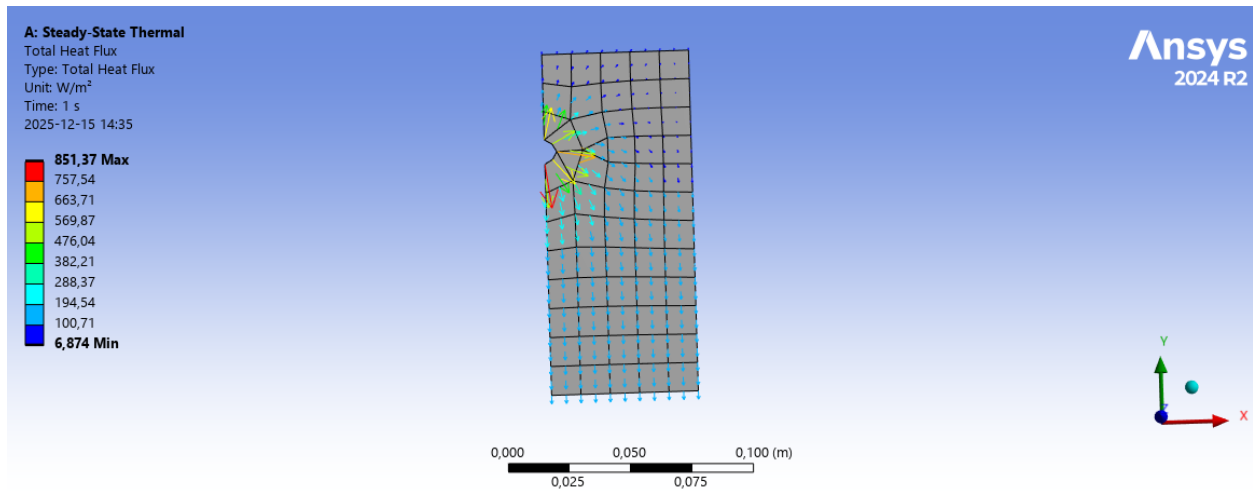


Figure 35: Direction of heat flow

5.2 SubTask

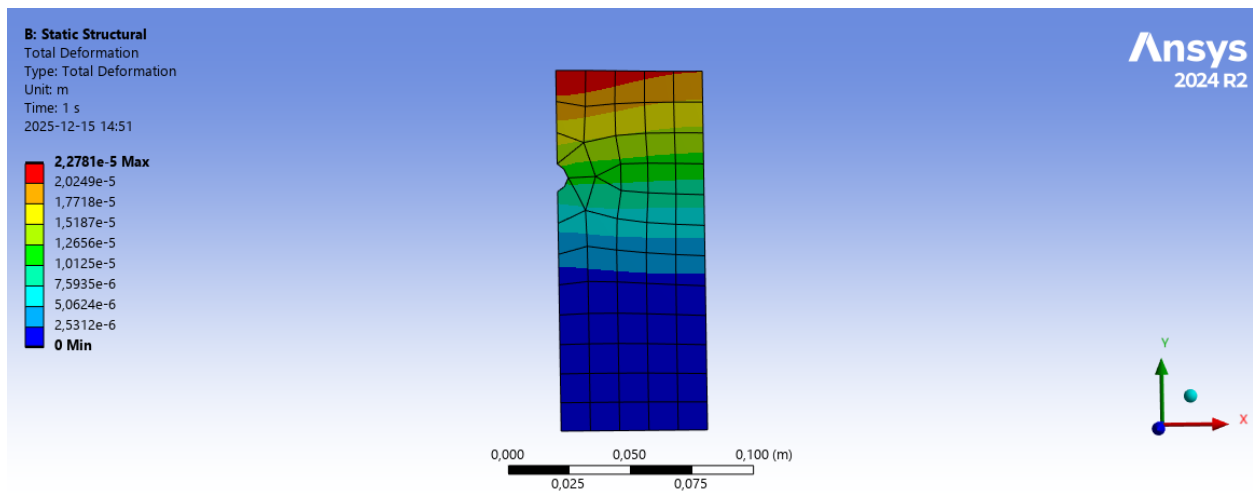


Figure 36: Total deformation

The total displacement of the concrete slab is 2,2781e-005 m.

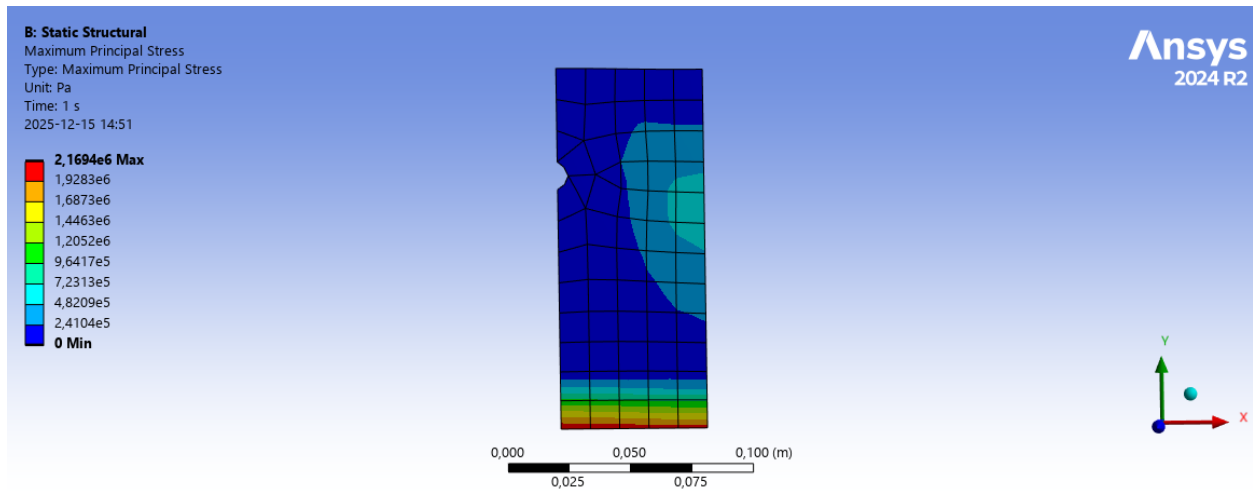


Figure 37: Maximum principal stress

The maximum principal stress obtained is 2,1694 MPa which is lower than the 3 MPa in tension.

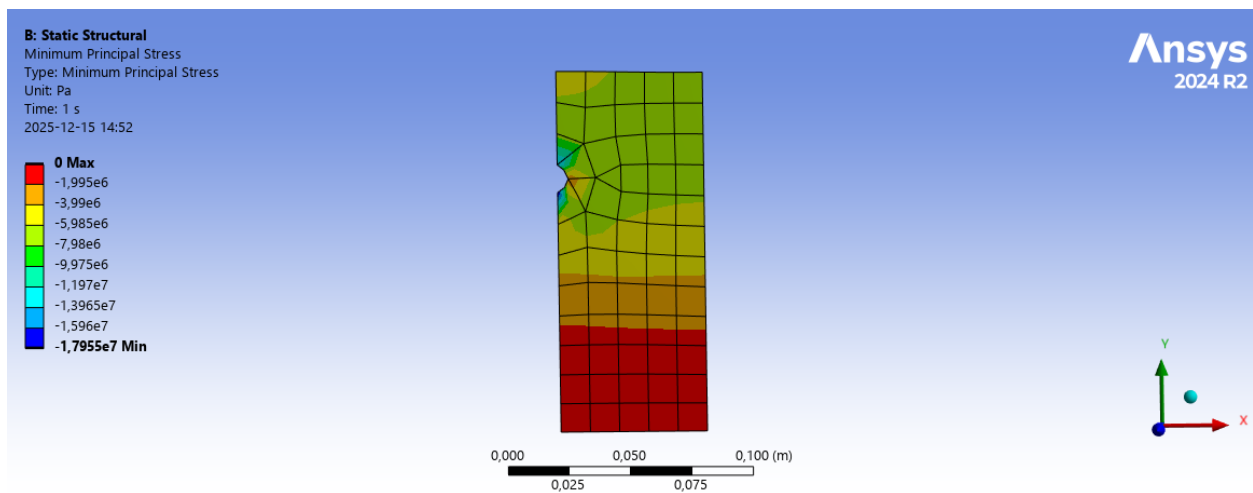


Figure 38: Minimum principal stress

The minimum principal stress obtained is 17,955 MPa which is less than the 30 MPa in compression.